

DATA ANALYTICS PROJECT REPORT

Prepared For:

Retail Business Management

Customer Sales Intelligence Dashboard

Prepared By	DataLens Analytics
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Confidentiality	Confidential — Client Use Only

Transforming Shopping Data into Revenue Decisions

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Executive Summary

Your retail business is sitting on a goldmine of customer data — and this report unlocks it. We analyzed 3,900 customer transactions to answer one question: where is revenue really coming from, and how can you get more of it? The analysis reveals that a specific customer profile mid-aged male shoppers drives disproportionate revenue, that your discount strategy is working against your margins in one critical category, and that over 3,000 loyal customers are currently underutilized as a growth engine.

Item	Details
Industry / Sector	Retail / Consumer E-commerce
Project Title	Customer Sales Intelligence Dashboard
Data Points Analyzed	3,900 customer purchase transactions
Tools Used	Python · PostgreSQL · Power BI
Primary Deliverable	Interactive Dashboard + Business Insights Report
Delivery Date	May 2026

Key Outcomes at a Glance

Mid-Aged Males (36–55) Highest Revenue Segment

Accessories Top Category by Orders

3,116 Loyal Customers Untapped Growth Pool

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About This Report

This report was prepared by DataLens Analytics as a deliverable for the Customer Sales Intelligence engagement. It presents a complete, data-driven analysis of customer purchasing behavior, revenue patterns, and product performance across your retail operation.

The report is structured to first establish the business context and core problem, walk through the analytical methodology applied, present key findings backed by data, and close with prioritized, actionable recommendations your team can implement immediately.

All findings are derived from 3,900 transactional records covering multi-season purchase data. Recommendations are based solely on the data analyzed and should be considered alongside operational knowledge held by your management team.

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Business Problem Statement

The retail company approached this engagement with the following core business challenge:

“How can we leverage consumer shopping data to identify which customer segments drive the most revenue, which product categories and channels to prioritize, and what marketing strategies will improve retention and profitability?”

The specific business concerns driving this engagement were:

- Uncertainty about which customer demographics generate the most revenue and how to target them effectively
- Lack of visibility into whether discount campaigns are helping or eroding average purchase value
- No clear view of which product categories and sales channels to invest in for maximum return
- An untapped loyal customer base with no structured engagement or retention strategy
- Inability to differentiate between subscriber and non-subscriber purchasing behavior

This project was scoped to answer these concerns using transactional data, transforming raw records into structured business intelligence that management can act on immediately.

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Project Scope & Objectives

Scope of Work

Included in Scope	Excluded from Scope
- Customer demographic revenue analysis - Product category performance analysis - Sales channel comparison (online vs. offline) - Discount impact on purchase value - Loyalty and repeat buyer behavior - Subscription vs. non-subscriber patterns - Shipping preference and spend correlation - Interactive Power BI dashboard	- Real-time data integration - Competitor benchmarking - Predictive ML / forecasting models - External market data analysis - CRM or ERP system integration

Project Objectives

#	Objective	Expected Deliverable
01	Identify highest-value customer segments	Revenue breakdown by gender, age group, and demographic profile
02	Evaluate discount impact on revenue	Category-level discount vs. spend analysis
03	Determine top-performing product categories	Category ranking by order volume, revenue, and satisfaction
04	Assess channel and shipping performance	Online vs. offline revenue split; express vs. standard shipping spend
05	Activate the loyal customer base	Loyalty segment profile and retention opportunity sizing

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Our Analytical Methodology

This engagement followed a structured, four-phase pipeline to ensure every finding is reliable, reproducible, and directly tied to a business outcome:

PHASE 1 Data Cleaning Python	PHASE 2 Exploratory Analysis Python / Jupyter	PHASE 3 Business Querying PostgreSQL	PHASE 4 Dashboard & Report Power BI
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Phase Details

Phase 1 — Data Collection & Cleaning: The raw dataset of 3,900 customer records was loaded into Python. We identified and resolved missing values, removed duplicates, corrected data types, standardized inconsistent categorical fields (gender, season, location), and engineered an Age Group column to enable demographic segmentation.

Phase 2 — Exploratory Data Analysis: Using Pandas, Matplotlib, and Seaborn, we explored revenue distributions by gender, age group, and location; purchase amount spread via histograms and box plots; category-level order volumes; seasonal purchase patterns; and review rating distributions correlated against purchase frequency.

Phase 3 — Business Querying: The cleaned dataset was loaded into PostgreSQL. Ten targeted SQL queries were written to answer specific business questions covering gender revenue, age group spending, category performance, discount impact, channel split, seasonal peaks, loyalty behavior, and shipping preferences.

Phase 4 — Dashboard & Reporting: An interactive Power BI dashboard was built featuring KPI cards, bar charts, donut charts, area charts and cross-filter slicers enabling management to explore the data independently by season, gender, category, and subscription.

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Dataset Overview

Data Summary

Attribute	Description
Data Source	Retail consumer purchase transaction records
Total Records	3,900 rows of customer transaction data
Data Format	CSV (raw) → Cleaned DataFrame → PostgreSQL tables
Key Fields Analyzed	Age, Gender, Purchase Amount, Product Category, Discount Applied, Review Rating, Season, Frequency of Purchase, Subscription Status, Shipping Type
Data Quality Issues Resolved	Missing values handled via imputation; duplicates removed; inconsistent category strings standardized; age groups engineered from raw age column

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Key Findings & Insights

Every finding below is derived directly from the data. Each is written as a specific, evidence-backed business statement not a vague observation with an assessed impact level to help you prioritize action.

#	Finding Area	What We Found	Impact
1	Gender Revenue Gap	Male customers generate higher total revenue than female customers across all product categories. Gender is the single most consistent revenue differentiator in the dataset.	HIGH
2	Male Discount Behavior	Male customers use discounts more frequently than females, yet their average purchase amount still exceeds the overall dataset average. This makes male customers discount-resilient — they respond to offers without sacrificing basket size.	HIGH
3	Top Value Segment	Mid-aged male customers (36–55) record the highest average spend of any demographic combination in the dataset. This segment is the single highest-value customer profile the business has.	HIGH
4	Category Satisfaction Leaders	Accessories and Footwear receive the highest customer review ratings of all product categories. These categories outperform on post-purchase satisfaction, indicating strong product-market fit.	MEDIUM
5	Subscription Revenue Paradox	Average spend per transaction is nearly identical between subscribers and non-subscribers. However, non-subscribers contribute greater total revenue — driven by	HIGH

		their significantly larger share of the customer base. Subscriptions are not driving premium spend.	
6	Clothing Discount Dependency	Clothing is purchased with a discount applied more frequently than any other category. Customers in this category appear price-sensitive and rely on promotional pricing to convert this erodes margin at scale.	HIGH
7	Loyal Customer Base Size	3,116 customers are classified as loyal buyers representing a substantial, identifiable retention base that currently receives no differentiated treatment or targeted engagement strategy.	HIGH
8	Accessories Order Leadership	Accessories lead all product categories in total order volume. Customers across all demographics purchase accessories more frequently than any other item, making it the highest-frequency revenue category.	MEDIUM
9	Repeat Buyer Profile	The majority of repeat buyers are non-subscribers. This is a structural gap: the customers who return most often are not enrolled in the subscription program, representing a direct missed-monetization opportunity.	HIGH
10	Youth Revenue Underperformance	Young customers (under 36) generate the lowest revenue of all age groups. This is consistent with lower disposable income and higher price sensitivity typical of this demographic.	MEDIUM
11	Mid-Aged Revenue Dominance	The 36–55 age group generates the highest overall revenue of all age segments. This cohort is the commercial backbone of the business across all product lines, not just premium categories.	HIGH
12	Express Shipping & Basket Value	Customers who select Express Shipping record a slightly higher average purchase amount compared to Standard Shipping users. Willingness to pay for speed correlates with higher overall spending intent.	MEDIUM
13	Universal Clothing Preference	Across every age group, Clothing is the highest-spend category, with Accessories consistently ranking second. This pattern holds regardless of demographic segment, confirming these two categories as the revenue backbone.	MEDIUM
14	Mid-Aged Clothing Revenue	The 36–55 age group drives the highest revenue specifically within the Clothing category, reinforcing that this cohort should be the primary target for all fashion and apparel campaigns.	HIGH

Impact Level: HIGH = Direct revenue or margin impact requiring immediate action. MEDIUM = Operational improvement opportunity with short-term ROI. LOW = Long-term strategic consideration.

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Business Recommendations

Priority 1 — Immediate Actions (0–30 Days)

Recommendation	Rationale & Expected Outcome
Launch a dedicated mid-aged male (36–55) campaign for Clothing	This segment spends the most in the highest-revenue category. A targeted campaign — even without discounts — will directly lift revenue from your most valuable cohort. Expected: increased AOV and category revenue.
Activate the 3,116 loyal customers with a retention program	These customers are identified but not engaged. A simple loyalty reward exclusive early access, points, or a personalized offer will improve retention rate and repeat purchase frequency. Expected: reduced churn, increased LTV.

Add Express Shipping as a checkout upsell prompt	Customers who choose express shipping spend more overall. Proactively surfacing this option at checkout will increase adoption and pull average basket value upward. Expected: marginal but consistent AOV lift.
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Priority 2 — Short-Term Actions (1–3 Months)

Recommendation	Rationale & Expected Outcome
Restructure Clothing discounts with minimum basket thresholds	Clothing is over-discounted, eroding margin without improving customer quality. Introduce a minimum spend to unlock discounts (e.g., discount applies on orders above \$40). Expected: protected margin, maintained conversion volume.
Convert repeat non-subscriber buyers into subscribers	Most repeat buyers are not subscribers, meaning your best customers are not in your most stable revenue model. Target them with a subscription trial offer highlighting benefits like free express shipping or exclusive access. Expected: recurring revenue growth.
Expand Accessories inventory and cross-sell with Clothing	Accessories lead in order volume and satisfaction ratings — they sell themselves. Bundling accessories with clothing purchases increases basket size at minimal acquisition cost. Expected: higher average order value across segments.

Priority 3 — Strategic Actions (3–6 Months)

Recommendation	Rationale & Expected Outcome
Build a youth acquisition strategy with entry-level price points	Young customers (under 36) currently underperform in revenue. They represent future mid-aged buyers. A low-cost entry strategy — affordable accessories, student pricing, or social-driven campaigns — builds this pipeline. Expected: long-term LTV growth.
Use Footwear and Accessories satisfaction as a referral engine	These categories have the highest review ratings, meaning customers are already delighted. A structured referral or review incentive program will convert satisfaction into new customer acquisition. Expected: organic growth at low cost.

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Tools & Technology Used

Tool / Technology	Type	Purpose in This Project
Python 3	Programming Language	Data ingestion, cleaning, feature engineering, and exploratory visualization
Pandas	Python Library	DataFrame operations, null handling, outlier detection, and age group segmentation
Matplotlib & Seaborn	Visualization Libraries	EDA charts: histograms, heatmaps, box plots, bar charts, and correlation matrices
Jupyter Notebook	Development Environment	Interactive analytical workspace for reproducible data preparation and EDA
PostgreSQL 15	Relational Database	Structured business querying — 10 targeted SQL queries answered all business questions
Power BI Desktop	BI & Visualization	Interactive dashboard with KPI cards, trend charts, demographic breakdowns, and cross-filter slicers

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Project Deliverables

#	Deliverable	Description	Format / Access
1	Cleaned Dataset	Fully cleaned 3,900-row dataset with engineered features, resolved quality issues, and standardized fields	.XLSX
2	SQL Query Bank	10 business-driven SQL queries covering all analytical objectives with inline comments	.SQL File
3	Power BI Dashboard	Interactive dashboard with 5+ visuals, DAX KPIs, and cross-filter slicers for self-service exploration	.PBIX File
4	This Insights Report	Complete business-focused report with findings, recommendations, and strategic roadmap	.PDF

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Value Delivered

The insights delivered through this engagement are estimated to support the following measurable business outcomes:

Business Outcome	How This Analysis Enables It
Increased revenue from top customer segment	Mid-aged male profile is now fully identified with spending behavior mapped — enabling precise campaign targeting without wasted budget on low-return segments
Improved margin on Clothing category	Discount dependency in Clothing is quantified, enabling management to implement threshold-based discounting that protects per-unit margin while maintaining conversion volume
Higher customer retention and lifetime value	3,116 loyal customers and the majority of repeat buyers are now a named, actionable cohort ready for structured loyalty and subscription conversion programs
Self-service business intelligence	The Power BI dashboard replaces ad-hoc reporting requests — any team member can now filter by season, category, or demographic and answer their own business questions in real time

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Next Steps

Recommended Client Actions

- Share findings with marketing and product teams and align on Priority 1 recommendations within the first two weeks
- Schedule a Power BI dashboard walkthrough session so your team can begin using it for day-to-day decisions
- Identify the 3,116 loyal customers in your CRM and assign them to a dedicated re-engagement campaign track
- Set a 30-day review to assess early results from the mid-aged male campaign and express shipping upsell test
- Implement minimum-spend thresholds on Clothing discounts and monitor margin change over 60 days

Potential Future Engagements

Service / Phase	Description
Phase 2: Customer Churn Prediction Model	Build a machine learning model to identify at-risk customers before they stop purchasing — enabling proactive retention at scale
Phase 3: RFM Segmentation & CLV Modelling	Advanced Recency-Frequency-Monetary segmentation to score every customer and predict lifetime value for budget allocation decisions
Monthly Dashboard Refresh	Ongoing data refresh and insight update service your dashboard stays current with new transaction data every month

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About the Analyst

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Data Analyst & Business Intelligence Specialist

I help retail and e-commerce businesses transform raw transactional data into clear, actionable intelligence using Python, SQL, and Power BI. I specialize in consumer behavior analysis, revenue segmentation, and sales intelligence dashboards that give management the clarity to make faster, more confident decisions — without needing a full-time data team.

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